

Double-Ended Queue (Deque) Introduction

Q. What is a Double-Ended Queue (Deque)? Explain its types and how it differs from a standard queue and stack. Provide real-world applications.

****Definition to Remember****

A ****Deque**** is a data structure that allows insertion and deletion of elements from both ends. It can perform operations like push, pop, push_front, pop_front, push_back, and pop_back. It is often used in algorithms where elements need to be added or removed from both ends of a sequence.

1. Operations in a Deque

Unlike standard queues (insert Rear, delete Front) or stacks (insert Top, delete Top), a Deque supports four primary operations:

1. Insertion

Visual Representation:

text

2. Types of Restricted Deques

To enforce specific behaviors, Deques are classified into two restricted variants:

Type	Restriction	Allowed Operations
1. Insertion	...	...
2. Deletion	...	...

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3. Comparison with Stack and Queue

Data Structure	Insertion Point	Deletion Point	Principle
Stack	Top	Top	LIFO
Queue	Rear	Front	FIFO
Deque	Both Ends	Both Ends	FIFO/LIFO

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4. Real-World Applications

* ****Web Browser History:**** When navigating back and forth, URLs are added/removed from one end. If history exceeds the max limit, oldest URLs are dropped from the other end.

* ****Undo/Redo:**** Similar to browser history, it uses a deque to manage a sequence of actions.

*****Must-Write Points (for 10 marks)*****

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*****Quick Recall*****

`Deque
History,